

Exercises from Ch. 14: Representation of One Endomorphism, Lang

Matthew Gergley

Last Updated: September 22, 2026

Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 4)

Show that the eigenvalues of the matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

in the complex numbers are $\pm 1, \pm i$.

Proof. Let

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Computing the characteristic polynomial,

$$\begin{aligned} P_M(t) &= \det(tI_4 - M) \\ &= \det \begin{pmatrix} t & 1 & 0 & 0 \\ 0 & t & 1 & 0 \\ 0 & 0 & t & 1 \\ 1 & 0 & 0 & t \end{pmatrix} \\ &= t^4 - 1. \end{aligned}$$

Over $\mathbb{C}$, we may write

$$P_M(t) = t^4 - 1 = (t - 1)(t + 1)(t - i)(t + i) \in \mathbb{C}[t].$$

Therefore, by Theorem 3.2, the eigenvalues of M are the roots of its characteristic polynomial and thus are $\pm 1, \pm i$. □

Exercise 0.2. (Exercise 11)

Let E be a finite-dimensional space over the field k . Let $\alpha \in k$. Let E_α be the subspace of E generated by all eigenvectors of a given endomorphism A of E , having α as an eigenvalue. Show that every non-zero element of E_α is an eigenvector of A having α as an eigenvalue.

Proof. Let $\alpha \in k$ and $A \in \text{End}_k(E)$. Given $v \in E_\alpha \subseteq E$, we may write

$$v = \sum_{i=1}^r a_i w_i, \quad a_i \in k$$

where w_i are the eigenvectors of A with eigenvalue α . Suppose $v \neq 0$. Then

$$\begin{aligned} Av &= A \left(\sum_{i=1}^r a_i w_i \right) \\ &= \sum_{i=1}^r a_i (Aw_i) \quad (\text{by linearity of } A) \\ &= \sum_{i=1}^r a_i (\alpha w_i) \\ &= \alpha \left(\sum_{i=1}^r a_i w_i \right) \\ &= \alpha v. \end{aligned}$$

Thus it follows that every non-zero element of E_α is an eigenvector of A with eigenvalue α . □

Exercise 0.3. (Exercise 12)

Let E be finite dimensional over the field k . Let $A \in \text{End}_k(E)$. Let v be an eigenvector for A . Let $B \in \text{End}_k(E)$ be such that $AB = BA$. Show that Bv is also an eigenvector for A (if $Bv \neq 0$), with the same eigenvalue.

Proof. Let $v \in E$ be an eigenvector for $A \in \text{End}_k(E)$ with eigenvalue $\lambda \in k$, that is, $Av = \lambda v$. Let $B \in Z(A) \subseteq \text{End}_k(E)$ (i.e. an element of the centralizer of A). Multiplying the relation $AB = BA$ on the right by v , we obtain

$$\begin{aligned} (AB)v &= (BA)v \\ &= B(Av) \\ &= B(\lambda v) \\ &= \lambda(Bv). \quad (\text{since } B \text{ is linear}) \end{aligned}$$

Hence, $A(Bv) = \lambda(Bv)$ which proves that $Bv \in E$ is an eigenvector, when $Bv \neq 0$, for A with eigenvalue $\lambda \in k$. □